MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMAT200
ACTUARIAL MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 hrs Maximum Marks: 70

Part A
Answer any FIVE questions. Each question carries 2 marks

1. Mean = $E(X)$ 1Mark
   Variance = $E(X^2) - E(X)^2$ 1Mark

2. For definition of pdf in two variables. 2Marks

3. This is a positive valued process $X(t)$ such that $Y(t)$, the logarithm of $X(t)$ is a Brownian motion with drift parameter μ and variance σ^2. pp290 2Marks

4. Time that the stochastic process first reaches a certain point. pp 297 2Mark

5. $Z = \sum_{k=1}^{\infty} b_k v(k+1) I_k$ pp 307 2Mark

6. $p_{i,j}^{\pi}(s,t) = \int_s^t \int_{r_1}^t p_{\overline{ii}}(s,r)\mu_{ik}(r_1)p_{\overline{kk}}(r_1,r_2)\mu_{ik}(r_2)p_{\overline{jj}}(r_2,t)dr_2dr_1$ pp316 2Mark

7. option price = $S(0)\phi(\alpha + \sigma\sqrt{N} - ke^{-\delta N})\phi(\alpha)$ pp362 2Marks

8. It is the difference between underlying assets price and the strike price. 2Marks

Part B
Answer any FIVE questions. Each question carries 6 marks
Maximum Marks from the section is 30

9. Probability distribution function of discrete r. v . 4Marks
   continuous r.v 3Marks

10. State j is said to be recurrent if $f_j = 1$. pp290 3Mark
    Absorbing state is a state from which there is no escape 3Marks

11. A stochastic process $N(t)$ that counts the number of such events that have occured up to time t. pp293 4Mark
    Any correct example 3Marks

12. If $t \leq s(B(t)|B(s) = b)$ is normal with mean b and variance (t-s). pp298 4Mark
    If $t < s, B(t)|B(b) = b$ is normal with mean rb and variance $r(s,t)$ where $r = t/s$ pp298 3 Marks

13. ${}_{k+1}V^{(i)} =_k V^(i) + \prod_k^{(i)} - \sum_{j\neq i}(p_{ij}(k,k+1))(b_k^{ij} +_{k+1} v^j -_{k+1} v^i)$. pp326 4Mark
    $\frac{d}{dt} = \delta_t \, {}_t v^{(i)} + \pi_i^{(i)} - \sum_{j\neq i} \mu_{ij}(t)(b_t^{(ij)} +_t v^t -_t v^{(i)})$ pp327 3Marks

14. $\frac{\partial}{\partial t}(p_{ij}(s,t)) = (p_{ij}(s,t))\mu_{ii}(t)$ 4Marks
    $(p_{ij}(s,t)) = k(s)e^{\int_s^t \mu_{ii} r dr}$ pp 314 3Marks

15. State any of the theorem with proof
    In an artitrage free financial market if there is a self - financing trading strategy for which $v(n)$ is a constant c for some n, then$c = V(0)(1+r)^n$
    or
    Financial market is arbitrage free iff $d < (1+r) < u$ 7 Marks

16. option price $= (1+r)^{-N} \sum_{m=0}^{N} {}^N c_m p^m (1-p)^{N-m}(S(0)u^m d^{N-m} - K)_+$ 7Mark

Part C
Answer any THREE questions. Each question carries 10 marks
Maximum Marks from the section is 30

17. Service table. 6Marks
    Employer sponsored pension plans 5Marks

18. (i) Stochastic process definition . 6Marks
Explanation 5Mark

19. (1) (i)$X(0) = 0$ (ii) the process has independent and stationary increments (iii) for each $t > 0$, $X(t)$ is normally distributed with mean 0 and variance $\sigma^2 t$ (iv) the relations $t- > x_t$ are continuous function of t. A standard B.M is one in which $\sigma^2 = 1$ pp296 4Marks
(2) Poisson - expectation propotional to the length of the interval. relations are step functions Brownian motion - normal distribution with variance propotional to the length of interval. Continuous relations 5Marks

20. (i) W_n the time between the $(n-1)^{th}$ event and n^{th} event pp295 5Marks
(ii) the probability is $e^{-\lambda w}$ 6Marks

21. (i) $\overline{A_\pi} \frac{v_0 v_1 \ldots v_M}{(\mu_0+\delta)(\mu_1+\delta)\ldots(\mu_{M-1}+\delta)}$ pp324 5Marks
(ii) $p_{02}(0,n) = \int_0^n p_{00}(s)\mu_{02}(s)p_{22}(s,n)ds$ 6Marks
$p_{02}(0,n) = \mu_1 \dfrac{e^{-(\mu_2+\rho+\epsilon_2)} - e^{-(\mu_1+\mu_2+\rho)n}}{\mu_1 - \epsilon_2}$ 6Marks

22. (a) No arbitrage required. $d < 1 + r < u$. Here $0.5 < 1.1 < 1.5$ 6Marks
(b) $S_u = 75, S_d = 25, p_u = 0, p_d = 25$ $p^* = \dfrac{1+r-d}{u-d} = 0.6$ 4Marks
Expected payoff $p_0 = 9.0909$ 2Marks

N.B Suitable stage marks may be fixed by the examiner for alternate correct answers.